Lautaro Casella

28 YEARS OLD
GAME PROGRAMMER | C# & UNITY

Profile

Software Developer and Video Game Programming student focused on mechanics creation, artificial intelligence, systems simulation and optimization. Solid knowledge of object-oriented programming (C#) and application of design patterns to build scalable architectures in Unity. Experience writing custom collision algorithms, dynamic weapon systems, and optimizing performance and memory in high entity-density environments.

Contact Info

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Location
Hurlingham, Buenos Aires, AR

Portfolio
lmcasella.github.io/portfolio

LinkedIn
linkedin.com/in/lautaro-casella-b990a6216/

Github
github.com/lmcasella

Technologies

Languages: C#, JavaScript, TypeScript, SQL.

Engines & Tools: Unity, Godot, Pixi.js, Git.

Game Architecture: Scriptable Objects, Strategy Pattern, Event-Based Architecture, Spatial Hashing (2D Optimization).

Software Engineering: Clean Architecture, SOLID, Data Structures, SCRUM Methodologies.

Featured Projects

Into the Unknown | Unity, C#
Link Itch.io | Repository

2D top-down action roguelike game featuring dynamic room generation, multiple combat abilities, and state persistence.

- **Architecture & SOLID:** Built modular component systems strictly interconnected through the Observer Pattern (Event-Based Architecture) to eradicate coupling between combat logic, progression, and the user interface.
- **Artificial Intelligence:** Programmed AI for various enemy types (melee, ranged, bosses) implementing Steering Behaviors for fluid pursuit and evasion movements.
- **Global Systems:** Implemented persistent managers to control the Game Loop, audio engine, and player stats conservation across level transitions.

Star Attack | Unity, C#
Link Itch.io | Repository

2D shoot 'em up arcade game featuring a wave-based progression system, dynamic difficulty scaling, and score management.

- **Modular Weapon Design:** Implemented the Strategy Pattern using Scriptable Objects to decouple shooting logic, allowing designers to create and balance new weapons directly from the Inspector without modifying the source code.
- **Optimization & Memory Management:** Developed collision detection systems and automatic entity cleaning (projectiles and enemy ships) upon leaving camera bounds to prevent memory leaks and framerate drops.

Languages

English: B2 (Currently studying)

- **Flow Control:** Built a centralized GameManager responsible for managing match state machines, controlling wave transitions, global timers, and the persistence of win/loss conditions.

Cyber Survivors | Pixi.js, JavaScript
Link Itch.io | Repository

2D bullet-heaven survival game focused on massive horde management and browser performance optimization.

- **Algorithmic Optimization (Spatial Hashing):** Implemented a spatial grid from scratch for collision detection, eradicating the computational cost of "all-vs-all" checks. This allowed processing and rendering hundreds of simultaneous entities without framerate drops.
- **State Machine Pattern:** Decoupled behavior logic and spritesheet control using State Machines, ensuring clean and scalable transitions.

Education

Universidad Nacional de Hurlingham
03/2024 - present

Bachelor's Degree in Video Game Programming and Simulations.

Universidad Nacional de Hurlingham
03/2018 - 12/2023 (graduated)

University Degree in Computer Science.

Professional Experience

Full Stack Developer
08/2023 - present • Municipality of San Martín

- Developed and maintained backend architecture in C# (.NET) and optimized database performance (SQL Server).
- Implemented robust security measures using token-based validation, and ensured secure integration with third-party government APIs.

Full Stack Developer
10/2021 - 12/2022 • Municipality of Hurlingham

- Engineered end-to-end internal systems and user authentication flows (OTP validation) using Angular and TypeScript.
- Participated in the full software development lifecycle operating under SCRUM agile methodologies.